UNIT 1 · KINEMATICS · BEFORE THE CART LAB

Mini Quiz — Kinematics Readiness

Physics 2026–27 · 20 minutes · 20 points · in class Tuesday, October 6

NAME _____ DATE _____

Three problems, about 20 minutes. These are Problem Set 5 problems with new numbers: run KDEA on each, and name the *missing variable* in your E row — a right answer with no E row does not earn full credit. The four equations are printed below; choosing between them is the quiz. No notes; a calculator is fine. Units the whole way.

$v_f = v_i + at$ · $\Delta x = v_i t + \tfrac{1}{2}at^2$ · $v_f^2 = v_i^2 + 2a\Delta x$ · $\Delta x = \tfrac{1}{2}(v_i + v_f)t$

1 The cart on the ramp

KDEA · 7 POINTS

A lab cart is released from rest at the top of a ramp and rolls with a constant acceleration of $0.50\ \text{m/s}^2$.

a. How long does it take to reach the $1.0\ \text{m}$ mark?

b. How fast is it moving as it passes that mark?

c. Where was the cart at *half* that time? Compute it, then say what fraction of the 1.0 m that is.

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

2 No time given

KDEA · 7 POINTS

A cyclist moving at $12\ m/s$ brakes with constant acceleration and stops in $18\ m$.

a. Find the acceleration, sign included.

b. How long does the stop take?

c. One sentence: how do the signs of v and a tell you the cyclist is slowing down?

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3 Two-stage trip

MULTI-STEP · 6 POINTS

A scooter starts from rest, accelerates at $2.0\ m/s^2$ for $5.0\ s$, then holds that velocity for another $6.0\ s$.

a. How fast is it moving at the end of the first stage?

b. How far does it travel during the first stage?

c. What is the total distance for the whole trip?

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Formative check — you may retake a similar version once (syllabus quiz policy). The cart lab starts right after this; Problem 1 is the experiment you are about to run.